

Barometric Pressure & Altitude Sensor (BMP280)

Hardware Overview

What is it?

The **BMP280** is a high-precision digital sensor designed to measure Barometric Pressure and Temperature. Because air pressure changes predictably with elevation, this sensor is famously used to calculate Altitude (height above sea level).

Electrical Explanation

Inside the chip is a tiny "piezoresistive" diaphragm that bends slightly under the weight of the atmosphere.

- Pressure: It measures the **Absolute Pressure** of the air around it in Pascals (Pa).
- Temperature: It includes a temperature sensor needed to compensate for the pressure readings (cold air is denser than hot air).
- Digital Precision: Like the **AHT10**, it processes data internally and sends a digital signal, making it immune to electrical noise.

Key Hardware Facts

- **Connection**: I²C Bus (SDA/SCL).
- **Default Address**: 0x77.
- **Logic Voltage**: 3.3V or 5V
- **Signal Flow**: Environmental Air → BMP280 Diaphragm → ADC → I²C Data → Processor.

Software & Programming

kcomp_BMP280.h

This library handles the mathematical compensation formulas required to convert raw sensor data into readable units.

Usage: `#include <kcomp_BMP280.h>`

Function Reference

Function Name	Description & Parameters
<code>bool initBMP280()</code>	Setup: Starts I ² C and reads calibration data. ▪ Returns <code>true</code> if sensor is connected, <code>false</code> if missing.
<code>float getBMP280Temperature()</code>	Read Temp: Returns temperature in °C.
<code>float getBMP280Pressure()</code>	Read Pressure: Returns absolute pressure in Pa. <i>Divide by 100 to get hPa (hectopascals).</i>
<code>float getBMP280Altitude(float seaLevel)</code>	Read Altitude: Calculates height in m. ▪ <code>seaLevel</code>: Pressure at sea level (usually ~1013.25 hPa).

⚠ Pitfalls & Troubleshooting - My Altitude is Wrong! (The Weather Problem)

You might find your altitude drifts even if the sensor is sitting still on your desk.

- The Reason: The sensor calculates altitude based on air pressure. However, the weather also changes air pressure.
 - Sunny day (high pressure): The sensor thinks it is *lower* (closer to sea level).
 - Stormy day (low pressure): The sensor thinks it is *higher* (up a mountain).
- The Math: The formula relies on a constant "Sea Level Pressure", usually about 1013.25 hPa. If the pressure changes by 1 hPa, the altitude changes by ~8.5 m.
- The Fix: For accurate absolute altitude, you must update the `seaLevel` parameter in `getBMP280Altitude()` with the current pressure reported by a local weather station.

Sample Program

This program initializes the sensor and prints weather data to the Serial Monitor.

```
#include <kcomp_BMP280.h>

void setup() {
    Serial.begin(115200);

    // 1. Initialize the sensor hardware
    if (!initBMP280()) {
        Serial.println("Error: BMP280 not found.");
        while(1); // Stop here
    }
    Serial.println("BMP280 System Online!");
}

void loop() {
    // 2. Read Values
    float temp = getBMP280Temperature();
    float press = getBMP280Pressure();

    // To get accurate altitude, update 1013.25 with the actual pressure
    float alt = getBMP280Altitude(1013.25);

    // 3. Output results
    Serial.print("Temp: ");
    Serial.print(temp);
    Serial.println(" *C");

    Serial.print("Pressure: ");
    Serial.print(press / 100.0); // Convert Pa to hPa
    Serial.println(" hPa");

    Serial.print("Approx Alt: ");
    Serial.print(alt);
    Serial.println(" m");

    // Wait 2 seconds
    delay(2000);
}
```

💡 **Try it out:** You can find the full sample program **TemperaturePressureBMP280** in the K-Comp examples folder!